One-pot PCR (head-tailed gene-region primers + barcoded-head primers, one tube)

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v1.2, 2026-10-21; strain amplicons, single- or multi-locus (validated sets); 50 µL reaction, KAPA HiFi HotStart

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Source: lab sheet *ONE POT PCR.xlsx*. Product = *barcode – head – gene region – head – barcode*, same as the two-step route. One tube per sample, one barcode per tube. Set for ~3–4 kb amplicons.

Use when: amplifying loci from **strains** (pure-culture DNA, Culture# → MolWork#), either a **single locus** or **several loci** per strain (e.g. ITS, LSU, TEF1, RPB2): one tube per strain × locus, one barcode per tube. Record locus and barcode for every tube in LIMS Stage 3b. **Not for** inhibitor-rich or low-copy templates: there is no 1st-PCR product to check.

Several loci in one tube (optional): gene-region primer pairs for more than one locus may be combined in the same reaction **only for a primer set that has passed both tests** on this template type: (1) the loci **co-amplify** in one tube (no dropout, no primer-dimer) and (2) their products **differ enough in length to show as separate bands** on the 1% check gel. Untested combinations: one locus per tube. The barcode still marks the strain (one barcode per tube); the locus is told apart by band size on the gel and by sequence after sequencing. Record the locus set for the tube in LIMS Stage 3b.

Reaction mixture (per 50 µL)

Master mix = ReadyMix + gene-region primers + water, 1.2X for *n* + 1 reactions (negative control included). Add barcode pair and template per tube.

Component	Volume	Final
2X KAPA HiFi HotStart ReadyMix	**25 µL**	1X
Gene-region primer F, head-tailed (10 µM)	**0.5 µL**	**0.1 µM**
Gene-region primer R, head-tailed (10 µM)	**0.5 µL**	**0.1 µM**
DEPC-treated water	**19.5 µL (minus 1 µL per extra locus)**	
Master mix subtotal	***45.5 µL***	
Barcoded-head primers F+R (10 µM each)	**1.5 µL**	**0.3 µM each**
DNA template	**3 µL**	
Total	**50 µL**	

Note Multi-locus tube (validated set only): add **0.5 µL** of each extra head-tailed F and R primer (0.1 µM each) and take the same volume off the water, so the total stays 50 µL.

Procedure

SETUP

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1. Clean bench with 1% bleach. Thaw reagents on ice, vortex, spin down.

2. Make master mix. Mix by pipetting, spin down.

3. Dispense **45.5 µL** master mix per tube.

4. Add **1.5 µL** barcoded-head primer pair, **one barcode per tube**. Write the barcode on the tube and in the

LIMS row.

5. Add **3 µL** template. Negative control: **3 µL** water.

6. Tap to mix, spin down, keep on ice.

CYCLING PROGRAM

7. Phase 1 anneals at the gene-region primer Ta; phase 2 at **65 °C** (barcoded-head primers).

Stage	Step	Temp	Time	Cycles
1	Initial denaturation	**98 °C**	**45 s**	**1×**
2	Denaturation	**98 °C**	**10 s**	**25×**
2	Annealing	**60–66 °C** (a)	**30 s**	
2	Extension	**72 °C**	**2 min 30 s (b)**	
3	Intermediate extension	**72 °C**	**5 min**	**1×**
4	Denaturation	**98 °C**	**30 s**	**1×**
5	Denaturation	**98 °C**	**15 s**	**15×**
5	Annealing	**65 °C**	**15 s**	
5	Extension	**72 °C**	**30 s**	
6	Final extension	**72 °C**	**2 min 30 s**	**1×**
7	Hold	**4 °C**	∞	

(a) From the primer list; LIMS method = one_pot_60 / 62 / 64. (b) ~1 min per kb.

Note 40 cycles total. Do not add phase-2 cycles (chimeras, primer dimers).

QC AND REGISTRATION

8. Gel: **2 µL** per lane, 1% agarose. One band at the expected size (multi-locus tube: one band per locus, each at its expected size); negative control blank. Head-only extra band → phase 2 incomplete. Band in negative control → discard batch.

9. Photograph gel, mark wells, upload to lab Google Drive, record pcr_ok.

10. Qubit Flex (dsDNA BR) each tube; enter concentration in LIMS.

11. LIMS Stage 3b (/lims-register): one pcr_reaction row per MolWork#, method = one_pot_60 / 62 / 64, taq = KAPA HiFi HotStart. Assign flow cell + barcode at setup; the single OK makes the sample ready for Stage 4a.

12. Pool equal molar → *Size selection (1X)* clean-up → EvoPrep library.

Materials

#	Material
1	2X KAPA HiFi HotStart ReadyMix
2	Head-tailed gene-region primers F + R, 10 µM (same as two-step 1st PCR)
3	Pre-mixed barcoded-head primer pairs, 10 µM each (barcode plate)
4	DEPC-treated water
5	DNA template, 1–10 ng/µL, MolWork# assigned
6	8-strip PCR tubes / 96-well plate, filter tips, 1% bleach

Note Barcoded-head primers must carry the same head sequence as the gene-region primer tails.